

OCR Further Mathematics
Mechanics - Impulse and Momentum
Mark Scheme
AS and A Level
Further Mathematics A - H235, H245



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Question		Answer/Indicative content	Marks	Part marks and guidance	
	b	Total initial KE = $\frac{1}{2} \times 2 \times 5^2 + \frac{1}{2} \times 6 \times 4^2 = 73$ $u_{Ay} = u_{Ay}, v_{By} = v_{By} = 2\sqrt{3}$ $v_{Ax} = -3$ KE Loss = $73 - \left(\frac{1}{2} \times 2 \times (3^2 + 4^2) + \frac{1}{2} \times 6 \times (2\sqrt{3})^2 \right) = 12 \text{ J}$	B1 (AO1.1) M1 (AO3.4) M1 (AO3.4) A1 (AO1.1) [4]	Perpendicular component found and unchanged. Using their formula for v_{Ax} from above part. NB If method mark for conservation of momentum not seen in part above then award M1 in the above part if either $m_A \times 3 + m_B \times -2 = m_A v_{Ax}$ or $2 \times 3 + 6 \times -2 = 2v_{Ax}$ seen here. If method mark for restitution not seen in part above then award M1 in above part if seen here.	
		Total	10		



Question		Answer/Indicative content	Marks	Part marks and guidance		
2	a	$\mathbf{l} = m\mathbf{v} - m\mathbf{u} = 2(-3\mathbf{i} + \mathbf{j} - (5\mathbf{i} + 16\mathbf{j}))$ $= 2(-8\mathbf{i} - 15\mathbf{j})$ $I = 2\sqrt{(-8)^2 + (-15)^2}$ $= 2\sqrt{289} = 34$ $\cos\theta = \frac{\mathbf{l} \cdot \mathbf{i}}{ \mathbf{l} \mathbf{i} } = \frac{-16 \times 1}{34 \times 1}$ $\theta = \cos^{-1} \frac{-8}{17} = 118.1^\circ \text{ or } 2.06 \text{ rad}$	M1 (AO1.1) A1 (AO1.1) M1 (AO1.1) A1 (AO1.1) M1 (AO1.1) A1 (AO1.1) [6]	Correct use of formula (award if $m\mathbf{u} - m\mathbf{v}$) Allow $16\mathbf{i} + 30\mathbf{j}$ $o \sqrt{(-16)^2 + (-30)^2}$ r oe Attempting to use the dot product of $\mathbf{l}$ and $\mathbf{i}$ to find the required angle	or using the cosine rule on vectors $\mathbf{u}, \mathbf{v}, \mathbf{l}$ to reach $ \mathbf{l} = 34$ or use of ordinary trigonometry eg $\tan\theta = \frac{-30}{-16}$	
	b	$\text{Init KE} = \frac{1}{2} \times 2 \times (5^2 + 16^2)$ $\text{Final KE} = \frac{1}{2} \times 2 \times ((-3)^2 + 1^2)$ $\text{Loss} = 281 - 10 = 271 \text{ J}$	M1 (AO1.1) M1 (AO1.1) A1 (AO1.1) [3]	281 J 10 J		
		Total	9			



Question		Answer/Indicative content	Marks	Part marks and guidance		
3	a	$u_{Ay} = \sqrt{3}$ or awrt 1.73 $v_{Ay} = u_{Ay} (= \sqrt{3})$ $\begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \cdot \begin{pmatrix} v_{Ax} \\ \sqrt{3} \end{pmatrix} = 0 \Rightarrow v_{Ax} = -3$ $3 \times 2\cos 60^\circ + 4 \times -5 = 3 \times -3 + 4v_{Bx}$ $v_{Bx} = -2$ $e = \frac{-2 - -3}{2\cos 60^\circ - -5}$ $e = \frac{1}{6}$ or awrt 0.17 Alternative method for last last 5 marks $3 \times 2\cos 60^\circ + 4 \times -5 = 3v_{Ax} + 4v_{Bx}$ $e = \frac{v_{Bx} - v_{Ax}}{2\cos 60^\circ - -5}$ $v_{Ax} = \frac{-17 - 24e}{7}$ $\begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \cdot \begin{pmatrix} \frac{-17 - 24e}{7} \\ \sqrt{3} \end{pmatrix} = 0$ $e = \frac{1}{6}$ or awrt 0.17	B1 (AO3.1b) B1 (AO3.1b) M1 (AO2.2a) M1 (AO3.1b) A1 (AO1.1) M1 (AO3.1b) A1 (AO1.1) M1 M1 A1 M1 A1 [7]	Correct per pendicular component of velocity of A before collision This component of A's velocity is unchanged by collision O $\tan 30^\circ = \frac{\sqrt{3}}{v_{Ax}}$ r Using perp endicularity of A's velocities to derive a value for v_{Ax} Conservati on of momentum with 4 terms Restitution Conservati on of momentum Restitution O	$v_A = 2\sqrt{3}$ se r en Could be positive if shown in diagram Allow one sign slip Could be positive Allow one sign slip $3v_{Ax} + 4v_{Bx} = -17$ $v_{Bx} - v_{Ax} = 6^e$ $v_{Bx} = \frac{-17 + 18e}{7}$	



Question			Answer/Indicative content	Marks	Part marks and guidance		
					$r^{\tan 30^\circ = \frac{\sqrt{3}}{\left(\frac{17+24e}{7}\right)}}$ Using perpendicularity of A's velocities.		
	b		$v_a^2 = (\sqrt{5})^2 + (-3)^2 \text{ so i}$ $\frac{1}{2} \times 4 \times 5^2 + \frac{1}{2} \times 3 \times 2^2$ $\frac{1}{2} \times 4 \times 2^2 + \frac{1}{2} \times 3 \times 12$ 56 or 26 56 – 26 = 30 J Alternative method for last 4 marks $\frac{1}{2} \times 4 \times 5^2 - \frac{1}{2} \times 4 \times 2^2$ $\frac{1}{2} \times 3 \times 2^2 - \frac{1}{2} \times 3 \times 12$ 42 or 12 so loss = 42 + (–12) = 30 J	M1 (AO3.1b) M1 (AO1.1) M1 (AO1.1) A1 (AO1.1) A1 (AO1.1) M1 M1 M1 A1 [5]	Attempt to find final speed (squared) of A vectorially Attempt to find total initial KE Attempt to find total final KE Either correctly calculated Not –30 Attempt to find KE change for B Attempt to find KE change for A Either correctly calculated and correctly used to find total energy change Not –30 Not –30	$v_A^2 = 12$ Can be implied by correct expressions and 30 seen. Can be implied by correct expressions and 30 seen.	

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Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	12	



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Question		Answer/Indicative content	Marks	Part marks and guidance		
4	a	$\cos \theta = \frac{1}{2}$ or $\sin \theta = \frac{\sqrt{3}}{2}$ $v \sin \theta = (1+m)V$ $V = \frac{\sqrt{3}v}{2(1+m)}$ $I = mV$ or $I = \pm(V - v \sin \theta)$ $V = \frac{v \times \frac{1}{2} \sqrt{3}}{1+m} \Rightarrow I = \frac{\sqrt{3}mv}{2(1+m)}$	B1(AO3.1b) M1(AO3.1b) A1(AO3.1b) A1(AO2.2a) [4]	Must be clear where it comes from ie. Shown in diagram or clear use of distances. May be other way round Conservation of momentum in direction of string Working must be seen AG Justification of using Impulse = change in momentum	θ is the angle between the string and the direction of AB when the string initially becomes taut	

Examiner's Comments

Candidates found it difficult to justify their thinking on this question. To be given the first mark candidates needed to find a trig ratio. As it is a 'show that' question the examiner needed to be certain that the candidate had found this from using the lengths of the string. Some candidates did not show detailed enough workings with a diagram to be given this mark. From this point candidates needed to use the conservation of momentum (this is easiest in the direction of the string)



Question			Answer/Indicative content	Marks	Part marks and guidance	
					to find an equation for the final velocity in terms of m and v. Most candidates were unable to show logical reasoning and seemed to use the answer given in the question to form their answer. For the final mark examiners had to be certain that the candidate had considered change of momentum on either particle. Just multiplying their expression for velocity by 'm' wasn't enough to demonstrate this. A significant amount of candidates were unable to access this question because they were trying to model the problem in 1 dimension.  AFL When answering a question (particularly 'show that') encourage candidates to write what they are using to set up the equation and in what direction to communicate clearly what the intention is. For instance: 'CoM parallel to the string' or 'Impulse on A'	



Question		Answer/Indicative content	Marks	Part marks and guidance		
	b	$KE = \frac{1}{2}(v^2 + (v \cos \theta)^2)$ $KE = \frac{1}{2} \left(\left(\frac{\sqrt{3}v}{2(1+m)} \right)^2 + \left(v \times \frac{1}{2} \right)^2 \right)$ $KE = \frac{v^2(4 + 2m + m^2)}{8(1+m)^2}$	M1(AO3.1b) M1(AO1.1) A1(AO1.1) [3]	Transverse component unchanged and using their longitudinal component Substituting in for V and $\cos \theta$ (could just be in speed equation) Or any equivalent single algebraic fraction	Condone consideration of speed squared or inclusion of m in KE	<u>Examiner's Comments</u> This was the most difficult question on the paper for candidates to access. To be given the first 2 marks candidates needed to identify the 2 components of B's final speed and square each of them. Out of the minority of candidates who managed this, the main error was to forget that $m = 1$ in the kinetic energy formula leading to an extra m in their final answer.

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Question			Answer/Indicative content	Marks	Part marks and guidance	
	c		$V = \frac{\sqrt{3}v}{2(1+m)} \rightarrow 0 \text{ as } m \rightarrow \infty$ B moves (approximately) in a circle around A	B1(AO3.2a) B1(AO2.2b) [2]	If m is very large then A is approximately stationary or B has only its transverse velocity of $\frac{1}{2}v$ after the string becomes taut <u>Examiner's Comments</u> Some candidates omitted this question after struggling with the first 2 parts; however this question was accessible if candidates had modelled the situation. Essentially if m is very large, then A will be stationary as B will be unable to move it. As A is a fixed point then B will travel in circular motion around it. Some candidates were able to infer circular motion but didn't refer to A being stationary. The most common incorrect answers were that B would stop or performing simple harmonic motion.	
			Total	9		



Question	Answer/Indicative content	Marks	Part marks and guidance		
5	A colliding with B: $2 \times 5u = 2v_A + 2v_B$ $\frac{v_B - v_A}{5u} = 1$ $v_A = -u$ and $v_B = 4u$ B colliding with C: $3 \times 4u = 3v_B + 5v_C$ $\frac{v_C - v_B}{4u} = e$ $v_B = \frac{u(3-5e)}{2}$ $e \leq 1 \Rightarrow v_B \geq \frac{u(3-5)}{2} = -u$ So A and B are both travelling in the negative direction, but $v_A \leq v_B$ hence B and A do not collide again, ie there are only two collisions (AG)	M1 (AO 1.1) M1 (AO 1.1) A1 (AO 1.1) M1 (AO 1.1) M1 (AO 1.1) A1 (AO 1.1) M1 (AO 2.1) E1 (AO 2.1a) [8]	Conservation of momentum NEL Both Conservation of momentum NEL Correct v_B oe; eg stating that the greatest leftwards speed of B occurs when $e = 1$ and evaluating this Conclusion correctly drawn	$v_C = \frac{3u(1+e)}{2}$, b ut but this need not be stated; ignore any wrong value seen, provided not used further Or could see eg assumption $v_B < v_A$ leading to $e < 1$ stated as a contradiction Allow statement such as 'B is not catching up with A' in place of a formal inequality,	



Question			Answer/Indicative content	Marks	Part marks and guidance		
						provided correct working is seen	
			Total	8			



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Question			Answer/Indicative content	Marks	Part marks and guidance		
6		a	$\frac{1}{2}mu^2 = mgh(1 - \cos \theta)$ $mu = 2mv_B$ $e = \frac{v_B}{u}$ $v_B = \frac{1}{2}u$ and $e = \frac{1}{2}$ $2m \times \frac{1}{2}u = mV_A + 2mV_B$ $\frac{1}{2} = \frac{V_A - V_B}{\frac{1}{2}u}$ $V_A = \frac{1}{2}u$ $\frac{1}{2}m\left(\frac{1}{2}u\right)^2 = mgh(1 - \cos \phi)$ $\frac{1}{8}mu^2 = \frac{1}{4}mgh(1 - \cos \theta) = mgh(1 - \cos \phi)$ $\Rightarrow \cos \phi = \frac{3 + \cos \theta}{4}$	M1 (AO 3.3) M1 (AO 3.4) M1 (AO 1.1) A1 (AO 1.1) M1 (AO 3.4) M1 (AO 3.4) A1 (AO 1.1) M1 (AO 3.4) A1 (AO 2.1) [9]	Conservation of energy for A Conservation of momentum Restitution Conservation of momentum for 2nd collision with unchanged v_B Restitution for 2nd collision Conservation of energy for A AG so elimination of $\frac{1}{2}mu^2$ oe must be clearly shown	v_B is velocity of B after first collision	



Question			Answer/Indicative content	Marks	Part marks and guidance		
		b	$V_B = \frac{1}{4}u$ $2m \times \frac{1}{4}u + m \times \left(-\frac{1}{2}u\right) = mV_A + 2mV_B$ $\frac{1}{2} = \frac{V_A - V_B}{\frac{1}{4}u - \left(-\frac{1}{2}u\right)}$ $V_B = -\frac{1}{8}u \text{ and } V_A = \frac{1}{4}u$ $\text{KE loss} = \frac{\frac{1}{2}m \times \left(\frac{1}{4}u\right)^2 + \frac{1}{2} \times 2m \times \left(-\frac{1}{8}u\right)^2}{\frac{1}{2}mu^2} = \frac{3}{32} = 9.4\%$ So percentage of KE remaining is 90.6%	B1 (AO 1.1) M1 (AO 3.4) M1 (AO 3.4) A1 (AO 1.1)	Conservation of momentum for 3rd collision Restitution for 3rd collision	Or with signs reversed, if different 'positive' direction chosen	
		c	During and between the 1st and 2nd collisions there is no external horizontal force acting on A and while there is an external force acting on B its velocity and hence momentum is the same just before the 2nd collision as it is just after the 1st collision	E1 (AO 2.4) [1]	Identifying the unchanged velocity for B in the two collisions		
		d	Between the 2nd and 3rd collisions a component of the tension in the string OA is horizontal and so the momentum of the system is not conserved	E1 (AO 2.4) [1]	Identifying the relevant external force acting		
			Total	16			



Question			Answer/Indicative content	Marks	Part marks and guidance		
7		a	Use of vector triangle for impulse and momentum $I^2 = 7.2^2 + 9.6^2 - 2 \times 7.2 \times 9.6 \cos 50^\circ$ $I = 7.43 \text{ (N s)}$ Alternative solution Change in momentum $\rightarrow = 7.2 \cos 50^\circ - 9.6$ $I^2 = (7.2 \cos 50^\circ - 9.6)^2 + (7.2 \sin 50^\circ)^2$ $I = 7.43 \text{ (N s)}$	M1 (AO 1.1a) M1 (AO 1.1) A1 (AO 1.1) M1 M1 A1 [3]	May be implied Use of cosine rule –4.971... (4.971...)² + (5.515...)²		
		b	$\frac{\sin 50^\circ}{7.43} = \frac{\sin \theta}{9.6} \text{ or } \frac{\sin 50^\circ}{7.43} = \frac{\sin \phi}{7.2}$ oe $\sin \theta = 0.9903 \text{ or } \sin \phi = 0.7428 \text{ oe}$ Angle with original direction is 132.0° Alternative solution $\tan \phi = \frac{5.515...}{4.971...} \text{ oe}$ oe $\phi = 48 \text{ oe}$ Angle with original direction is 132.0°	M1 (AO 1.1a) A1 (AO 1.1) A1 (AO 1.1) M1 A1 A1 [3]	e.g. use of cosine rule to find θ or ϕ $\cos \theta = 0.1386, \cos \phi = 0.6696$ Allow positive or negative value 46.967...		
			Total	6			



Question			Answer/Indicative content	Marks	Part marks and guidance		
8		a	Components of initial vel: $6.5 \cos 30^\circ \downarrow$, $6.5 \sin 30^\circ \rightarrow$ Upwards component of final vel: $v_V = \frac{2}{3} \times 6.5 \cos 30^\circ$ $-2 = 2.5v_H - 2.5 \times 6.5 \sin 30^\circ$ $v_H = 2.45$ $v = \sqrt{v_V^2 + v_H^2}$, $\tan \theta = \frac{v_H}{v_V}$, oe Velocity has magnitude 4.48 ms^{-1} at 33.1° to the normal	M1(AO3.1b) M1(AO1.1) M1(AO1.1) A1(AO1.1) M1(AO1.1) A1(AO3.2a) A1(AO3.2a) [7]	For correct use of either component Use of impulse-momentum ; must involve horizontal components only For use of either trig or Pythagoras	(3.7527...)	Condone 33.2°
		b	$\frac{\frac{1}{2} \times 2.5 \times (6.5^2 - 4.48^2)}{\frac{1}{2} \times 2.5 \times 6.5^2} \times 100$ Percentage loss is = 52.5	M1(AO1.1a) A1(AO1.1) [2]	Factor $\frac{1}{2} \times 6.5$ s of may be omitted Condone answers 52 and 53		
			Total	9			



Question			Answer/Indicative content	Marks	Part marks and guidance	
9		i	$4(1) + 2(-3) = -4v_A + 2v_B$ $-v_A - v_B = -e(1 - (-3))$ $v_A = \frac{1}{3}(1 + 4e)$ and $v_B = \frac{1}{3}(8e - 1)$ $v_B > 0 \Rightarrow e > \frac{1}{8}$	*M1 A1 *M1 A1 dep*M1 A1 A1 [7]	Attempt at use of conservation of linear momentum Attempt at use of restitution equation, must be correct way round Must be consistent with direction used for CoLM Solve simultaneous equations to find v_A or v_B AEF, may be unsimplified. AG Correctly shown from correct v_B Examiner's Comments This question was a good source of marks for many candidates. A few made a sign error in one of the two equations so the directions were not consistent (the direction used as positive was not always clear and diagrams, if shown, frequently failed to indicate the direction of the spheres after collision). Most picked up the mark for solving although there were some who obtained expressions for e in terms of v_A or v_B from which they tried to show $e > 1/8$ without finding, as asked, the speeds of the spheres. The accuracy mark for the speeds was frequently lost because the chosen directions led to a correct velocity for v_A which was not converted to speed.	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	KE before $= \frac{1}{2}(4)(1)^2 + \frac{1}{2}(2)(3)^2$ $= \frac{1}{2}(4)\frac{(1+4e)^2}{9} + \frac{1}{2}(2)\frac{(8e-1)^2}{9}$ KE after $11 - \frac{2}{9}(1+4e)^2 - \frac{1}{9}(8e-1)^2 = \frac{5}{2}$ $e = \frac{7}{8}$	B1 B1ft M1 A1 [4]	11 J or $\frac{1}{2}(4)(1)^2 - \frac{1}{2}(4)\left(\frac{1+4e}{3}\right)^2$ Allow in terms of 1 variable eg v_A or v_B or e; $\frac{1}{2}(2)(3)^2 - \frac{1}{2}(2)\left(\frac{8e-1}{3}\right)^2$ Use KE before – KE after = ± 2.5 to get equation in 1 unknown AG Correctly shown Examiner's Comments Most got the first 3 marks of this part, though some got the masses mixed up, and others failed to follow through their speeds. Quite a lot claimed to get 7/8 when they clearly had not. A small number of candidates (generally who had got in a mess expressing speeds in terms of e) wrote the KE equation in terms of speed, substituted for one and solved, and found e correctly that way. A minority of candidates used an equation, $\frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} (u_A - u_B)(1 - e^2) ,$ that is not a requirement for this specification, but is an acceptable method. However, as with all formulae used, they must be quoted correctly.	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$2(-2) + 3(3) = 2w_B + 3w_C$ $w_B - w_C = 5f$ Use $w_B > \frac{3}{2}$ or $w_C > 0$ $f > \frac{1}{6}, f < \frac{1}{2}$	M1 M1 M1 A1A1 [5]	Attempt at use of conservation of linear momentum, cv(-2) Attempt at use of restitution equation, must be correct way round, cv(-2) Dependent on correct NEL and momentum equations with cv, correct cv (3/2) leading to $f > a$ where $0 < a \leq 1$; allow use of equality to get an inequality for f ; $w_B = 1 + 3f$ and $w_C = 1 - 2f$ Examiner's Comments Complete solutions to this part were very rare, with a significant number of candidates making no attempt at all. Those who did make an attempt usually gained the first 2 marks for a conservation of momentum equation and a restitution equation. Not all of those realised that they then had to use the fact that, after the second collision, B needed to move faster than A , and very few realised that they had to use the fact that the velocity of C remained positive.	
			Total	16		



Question			Answer/Indicative content	Marks	Part marks and guidance		
10		i	Momentum equation $2ma = -2m \frac{1}{10} \sqrt{5} \frac{1}{\sqrt{5}} + 7m \frac{1}{10} \sqrt{5} \frac{1}{\sqrt{5}}$ $(a =) \frac{1}{4} (\text{ms}^{-1})$ Comp of speed of A perp = 0.2 Speed of A was $\sqrt{(0.25^2 + 0.2^2)}$ OR $\tan \theta = \frac{0.2}{0.25}$ Speed 0.320 or $\frac{\sqrt{41}}{20}$; Ang 38.7° or 0.675 rads NLM $0.1 + 0.1 = -e(0 - a)$ (e =) 0.8	M1 A1 B1 M1 A1 M1 A1 [7]	Along line of centres Allow errors with signs and masses soi soi Allow their vel comps oe For both angle and speed Along line of centres Allow errors with signs Examiner's Comments There were many correct and concise answers to this part. There were also many answers where candidates wrote down everything they could about the situation, including considering momentum perpendicular to the line of centres, and were then not able to see a way through the problem. There were quite a number of cases of candidates omitting to find either the speed of A, or its direction of motion, or the coefficient of restitution, in an otherwise correct solution.	Allow use of 63.4° for full marks Must use comp of vel 0.320156; 38.6598° or 0.67474 rads May see	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	A and B have same speed perpendicular to line of centres after first collision	B1 [1]	accept 'vertical'	Examiner's Comments Some candidate answered this question accurately and briefly by stating that the (component of) speed of both A and B perpendicular to the line of centres remained equal. Other candidates filled up all the answer space, often with unnecessary explanations that there would be a collision if U is large enough – which was stated in the question. Many of these candidates rescued their answer by including the correct statement, in some form.

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Question			Answer/Indicative content	Marks	Part marks and guidance		
		iii	Momentum equation along line of centres $3mU - 2m\frac{1}{10}\sqrt{5}\frac{1}{\sqrt{5}} = 3mc + 2ma'$ NLM $a' - c = -1(-\frac{1}{10}\sqrt{5}\frac{1}{\sqrt{5}} - U)$ Use $a' = 0.11$ Max $U = \frac{1}{15}$	M1 A1 M1 A1 B1 A1 [6]	Allow errors with signs and masses Allow $\cos \alpha$ for $\frac{1}{\sqrt{5}}$ Allow errors with signs Accept any inequality Accept 0.0667 accept $\leq$ Examiner's Comments The most common successful approach to this part was to write down an equation for the conservation of momentum parallel to XY and another using Newton's experimental law, and then use the fact that as long as U is not greater than the speed of B there will be no second collision. Many candidates were unsuccessful on this part because of sign errors or using an incorrect mass in the momentum equation. A small number of candidates wrongly stated that U needed to be strictly less than $\frac{1}{15}$.	Must use comp of vel Or conservation of energy $\frac{1}{2}3mU^2 + \frac{1}{2}2m0.1^2 = \frac{1}{2}3mc^2 + \frac{1}{2}2ma'^2$ do not accept <	



Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	14	



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Question			Answer/Indicative content	Marks	Part marks and guidance		
11			Impulse / momentum triangle with sides 0.8, 1.2 and 1	B1	OR $1.2 \cos \alpha = \cos \theta - 0.8$		
			$\cos \theta = \frac{0.8^2 + 1^2 - 1.2^2}{2 \times 0.8 \times 1}$	M1	$1.2 \sin \alpha = \sin \theta$ M1 $1.44 = (\cos \theta - 0.8)^2 + \sin^2 \theta$ A1	Square and add	
			82.8° or 1.44 rads	A1		82.81924°	
			$1.2 \sin \alpha = \sin \theta$	M1	isw	or 1.445 rads	
			Angle 124°	A1	cv θ ; OR from cos rule	may see 55.771° or 0.97339 rads	
				[5]	No isw	2.168 rads	
					Examiner's Comments		
				Most candidates correctly applied the cosine rule, with or without a diagram, to find the angle between the directions of motion before and after the impulse acts. Most then went on to use the sine rule to find the angle between the direction in which the impulse acts and the initial direction of motion of the particle, though many wrongly gave the answer as 55.8°.			
				A small number of candidates drew an incorrect triangle of impulse / momentum. More candidates than usual attempted the question by looking at the motion parallel and perpendicular to the original direction of motion of the particle. A considerable number of these did no more than write down the initial equations.			

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Question			Answer/Indicative content	Marks	Part marks and guidance		
12		a	$2.5u \cos \theta - 3u \cos \theta = -2.5v_1 \cos \alpha + 3v_2 \cos \beta$ $-v_1 \cos \alpha - v_2 \cos \beta = -e(u \cos \theta + u \cos \theta)$ $u \sin \theta = v_1 \sin \alpha \text{ and } u \sin \theta = v_2 \sin \beta$ $\tan \beta = \frac{11 \tan \theta}{10e - 1}$	*M1(AO3.4) A1(AO1.1) *M1(AO3.4) A1(AO1.1) *B2(AO1.1a,1.1a) dep *M1(AO3.1b) E1(AO2.1) [8]	Attempt at use of conservation of linear momentum Attempt at use of restitution equation, must be correct way round Must be consistent with the directions used for conservation of linear momentum B1 for one correct Solving simultaneous equations by eliminating v_1 and v_2 use of $\tan X = \frac{\sin X}{\cos X}$ www; AG		



Question			Answer/Indicative content	Marks	Part marks and guidance		
		b	$10e - 1 = 0 \Rightarrow e = \frac{1}{10}$ $\tan \alpha = \frac{11 \tan \theta}{1 + 12e}$ $\tan 50 = \frac{11 \tan \theta}{1 + 12/10}$ $\theta = 13.4^\circ$	*M1(AO2.2a) B1(AO3.1b) dep*M1(AO3.1b) A1(AO1.1) [4]	Setting the denominator of $\tan \beta$ equal to 0 and obtaining a value for e Substituting their e and $\alpha = 50^\circ$ and solving for θ 13.40634... BC		
			Total	12			
13		a	$I = \int t^2 + 3e^t dt = \frac{1}{3}t^3 + 3e^t$ $I = \int_0^4 F dt = \frac{55}{3} + 3e^4 = 182$, so impulse is 182 N s	M1(AO3.3) A1(AO1.1) A1(AO1.1) [3]	Integrating one term of F correctly BC	182.1277...	
		b	$2(v - 5) = \frac{55}{3} + 3e^4$ Speed is 96.1 m s ⁻¹	M1(AO3.4) A1(AO1.1) [2]	Use of $I = mv - mu$	96.0638...	
			Total	5			



Question			Answer/Indicative content	Marks	Part marks and guidance	
14		i		M1	Momentum for Q/R , no g , at least 3 correct terms	
		i	$0.3u + 0.6 \times 0.8 = (0.3 + 0.6) \times 1$	A1	NB 0.48 in “before” from 0.8×0.6 ; not $1.5 \times 0.1 + 1.1 \times 0.3$ (A0)	
		i	$u = 1.4 \text{ m s}^{-1}$	A1		
		ii	$0.1 \times 1.5 + 0.3 \times 1.1 = \pm 0.1v + 0.3 \times 1.4$	M1	P, Q +ve “before”, allow P –ve “after”. Accept cv (1.4)	
		ii	$v = 0.6$	A1	Velocity of P , will be –ve if $-0.1v$ in momentum equation, accept $v = \pm 0.6$	
		ii	Momentum change = $\pm 0.09 \text{ kg m s}^{-1}$	A1	Tolerate loss of – sign if “small – large” has +ve answer	
		ii	OR			
		ii	Momentum change $Q = \pm 0.3(1.4 - 1.1) = \pm 0.09$	M1A1	Change for P is the change for Q	
		ii	Momentum change $P = \pm 0.09$	A1		
		ii	OR			
		ii	$0.1 \times 1.5 + 0.3 \times 1.1 + 0.6 \times 0.8 = (\pm)0.1v + 0.9(\times 1)$	M1A1	Overall equation	
		ii	Momentum change $P = \pm 0.09$	A1	From $\pm (0.9 \times 1 - 0.3 \times 1.1 - 0.6 \times 0.8)$	
					Examiner's Comments	
					Though the configuration and motion of the particles was elementary, it was necessary for candidates to consider the second collision before the first. Only a few avoided starting with the first collision, but nearly all corrected their approach after one or two lines of work.	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
					Possibly the commonest source of error was the creation of equations in which candidates multiplied two speeds together, having lost track of which numbers were masses, which velocities. Many candidates, who had found the correct value of the speed of Q in part (i), realised when doing part (ii) that there was no reason to work out the “after” speed of P, but that the answer could be found immediately from “after” momentum of the particle. Inclusion of P is automatically M0. Inclusion of 0.1 (unless a MR) is automatically M0 even in the case $0.1 \times 1.5 + 0.3u + 0.6 \times 0.8 = 0.1 \times 1.5 + (0.3 + 0.6) \times 1$ $u = 1.4$ Exceptions: use of 0.6 or $v(ii)$ instead of 1.5 in the example above is valid part (ii) can be done first, then $0.1 \times 0.6 + 0.3u = 0.1 \times 1.5 + 0.3 \times 1.1$, $u = 1.4$	
Total				6		

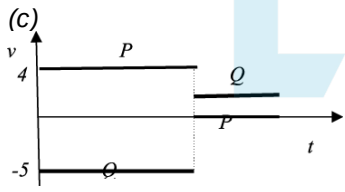
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Question			Answer/Indicative content	Marks	Part marks and guidance	
15		i		M1	Cons of momentum, no g^* , common v "after" term	
		i	$0.3 \times 4 - 0.2 \times 5 = +/-(0.3 + 0.2)v$	A1	$0.3 \times 4 + 0.2 \times 5 = +/-(0.3 + 0.2)v$ is M1A0A0	
		i	$v = 0.4 \text{ m s}^{-1}$	A1	Must be positive *Allow g if fully cancelled in first line BOD Examiner's Comments There were few incorrect solutions.	
		ii	(a) Q (or P at rest)	B1	If P moves, allow $0.3v$ when considering M1	
		ii		M1	Cons of momentum, no g^* , one "after" term	
		ii	$0.3 \times 4 - 0.2 \times 5 = 0.2v$	A1	$0.3 \times 4 + 0.2 \times 5 = 0.2v$ is M1A0A0	
		ii	$v = 1 \text{ m s}^{-1}$	A1		
		ii			*Allow g if fully cancelled in first line BOD Examiner's Comments Though this was an uncommon request, most scripts showed correct solutions. Some indicated faulty logic: "As P has the greater momentum, it will be moving after the collision". A significant minority of candidates believed P to be in motion after the collision, with speed 0.667 m s^{-1} .	
		ii	(b) $4t + 5t = 3.6$	M1	Or $9t = 3.6$, Or both $3.6 - x = 4t$ and $x = 5t$	
		ii	$t = 0.4$	A1		
		ii	$x_Q = 5 \times 0.4 (= 2)$	A1	Finds initial Q distance. $3.6 \times 5 / (4 + 5)$ is M1A1A1	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$T = (2/1 =) 2 \text{ s}$	A1		
		ii	OR			
		ii	(Time =)	M1	Equates pre-collision times	
		ii	$x/5 = (3.6 - x)/4$	A1	x is distance Q travels before collision	
					Examiner's Comments	
					This question took candidates away from formal specification topics (such as constant acceleration), and many were unsure how to proceed. There were however many correct answers gaining full marks.	
		ii	$x = 2 \text{ m}$	A1		
		ii	$T = 2/1 = 2 \text{ s}$	A1		
		ii	(c)	B1	One horizontal, +ve v intercept	
		ii		B1	One horizontal, -ve v intercept, terminates at same t	
		ii		B1	One along t -axis, starts at same t as +ve line ends, label P	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii		B1	One horizontal above t-axis, starts at same t as $-ve$ line ends. (Ignore any values put on graphs) Examiner's Comments Speed-time diagrams were fairly common, as was a failure to label which line segment related to which particle. Many diagrams showed P and Q with zero initial velocity, and graph lines which sloped.	
			Total	15		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
16		i	$0.3 \times 0.4 + I_x = 0.3 \times 0.6 \cos 30^\circ$	M1	$I_x = (\pm) 0.035884572$	Similar with velocities.
		i	$I_y = 0.3 \times 0.6 \sin 30^\circ$	M1	$I_y = (\pm) 0.09$	
		i	Magnitude of $I = 0.0969$ (Ns)	A1	CAO	0.096890
		i	68.3° from original direction	A1	Accept 111.74° if clear on diagram	68.26° . Direction must be clear in words or on diagram
		i	OR correct triangle of momentum	(M1)		Similar with velocities.
		i	$I^2 = 0.18^2 + 0.12^2 - 2 \times 0.18 \times 0.12 \times \cos 30^\circ$	(A1)	Allow for wrong angle	Or slip in figure
		i	$I = 0.096890$ (Ns)	(A1)	CAO	
		i	68.26° from original direction	(A1)	Accept 111.74° if clear on diag.	Examiner's Comments Most candidates drew a correct diagram for this situation and used the cosine rule to find the magnitude of the impulse. Those who then used the sine rule to find the angle usually failed to realise that the interior angle of the triangle was obtuse. Candidates who found components of the impulse, or used the cosine rule to find the angle, did not have any difficulty here. A small number of candidates did not get the momentum (or velocity) triangle correct.
		ii	Magnitude = 0.0969 N	B1ft		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	Direction opposite to (i) clearly shown and labelled or stated	B1ft	Accept direction at angle 141.7° with direction of 0.6 m s^{-1} motion	If total 0 scored, allow SC1 for 38.3° or 68.3° seen Examiner's Comments Most candidates either had this correct, or correct on follow through from part (i). There were some, however, who had quite different answers for both the magnitude of the second impulse and its direction.
			Total	6		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
17		i	C of M $2m \times 5 \cos \alpha - 3m \times 3\frac{1}{4} \beta = 2ma + 3mb$	M1*	allow sign and number slips.	a and b are vels of A and B to right
		i	$2mx4 - 3mx1.25 = 2ma + 3mb$	A1	$(2a + 3b = 4.25)$	
		i	Newton's experimental law	M1*	Or equivalent; allow sign slips	
		i	$b - a = \frac{2}{3}(-1.25 - 4)$	A1	$(b - a = 3.5)$	Consistent
		i	Attempt to solve simultaneous equations	*M1		
		i	$b = 2.25 \text{ (m s}^{-1}\text{)}$	A1	CAO	-2.25 if b defined to left
		i	$(a = -1.25 \text{ so) speed of } A = 3.25 \text{ (m s}^{-1}\text{)}$	A1	CAO	Examiner's Comments Many candidates gained full marks for this part, although some made a sign slip in one or other of the equations, some made an error in one or both of the masses and others made an error in solving the simultaneous equation. A small number of candidates failed to find the speed of A after finding the component of its velocity along the line of centres and a small number of candidates omitted to find the component of the velocity of B along the line of centres.
		ii	A and B both have same component of velocity perp to l o c	B1		
		ii	After collision with wall B must move faster than A	B1	May be implied	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	Coeff of restitution $> 5/9$ (accept > 0.5 recurring, or 0.556)	B1	Do not allow $\geq$	Ignore $e \leq 1$, etc Examiner's Comments While there were many good answers, a considerable number of candidates lost a mark by failing to point out that the speeds of A and B perpendicular to the line of centres were equal.
			Total	10		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
18		i	Impulse momentum diagram	M1	Right-angled triangle with angle α and sides labelled 3, v and $I/0.2$ or 0.6, $0.2v$ and I	Correct orientation, α and one side labelled correctly, right angle implied first two marks may be implied by correct working
		i		A1		
		i	$\tan \alpha = I/(0.2 \times 3)$	M1		
		i	$I = 0.25$ shown	A1	AG	
		i	OR $0.2 \times 3 = 0.2v \cos \alpha$	M1	Resolve parallel or perp to dir of motion	
		i	and $I = 0.2v \sin \alpha$	A1	Both	
		i	$\frac{I}{0.2 \times 3} = \tan \alpha$	M1	Attempt to manipulate	
		i	$I = 0.25$	A1	AG	
	Copyright © 2024 Exam Papers Practice	ii	$\cos \alpha = 3/v$	M1	Or using Pythagoras, with 3 and 1.25 oe	Examiner's Comments Almost all candidates did this part completely correctly, though not always by efficient methods. Some candidates found the answer to part (ii) on their way to the answer to this part.
		ii	(speed) = 3.25 m s^{-1}	A1	Examiner's Comments This part was almost always done correctly, though a small number of candidates ignored the vector nature of momentum and impulse and used $I = mv - mu$, with $I = 0.25$, $m = 0.2$ and $u = 3$.	



Question			Answer/Indicative content	Marks	Part marks and guidance	
			Total	6		
19				M1	Examiner's Comments	Use of cos rule; condone + for – / missing 2/ missing '0.6'; angle as 'θ' for M1
					Most candidates found this question straightforward, with about half using the cosine rule to find l and then the sine rule to find θ and about half finding expressions for $l \sin \theta$ and $l \cos \theta$ and solving these. A small number of candidates completed the diagram wrongly, and so ended up with wrong answers. Others made small errors, like omitting the mass from one or more term.	
			$l^2 = 2.04^2 + 0.9^2 - 2 \times 2.04 \times 0.9 \times \frac{15}{17}$	A1	And attempt to square root	Condone + for –
			1.32 (N)	A1	CAO	(1.3159)
				M1	Correct use of sin rule from their diagram oe	Can be in terms of l and θ
			46.8 (°) with initial direction of ball	A1	CAO	(46.8476) (0.8176 rads)
					OR $0.9 + l \cos \theta = 0.6 \times 3.4 \times \frac{15}{17}$ M1 $l \sin \theta = 0.6 \times 3.4 \times \frac{8}{17}$ M1 square and add to find l^2 ; or divide to find θ M1 l, θ A1 A1 CAO	Accept 46.7 from using $l = 1.32$ Allow missing 0.6 and / or sign or trig error for these 2 marks, then M0A0A0
			Total	5		



Question			Answer/Indicative content	Marks	Part marks and guidance	
20		i	Vel unchanged perp to L o C	M1	Examiner's Comments Almost all candidates did this question very well, working methodically through the three parts, with only an occasional sign error, or a confusion of sine and cosine. However, a small number of candidates thought that, after the collision, the motion of <i>B</i> was at right angles to the line of centres; the question clearly states that the direction is at right angles to <i>B</i> 's original direction of motion. These candidates were restricted to a method mark in each part.	Stated or used
		i	$0.6\sin 30^\circ = v\cos 30^\circ$	M1		Allow 1 sign or trig error
		i	$0.2\sqrt{3} \text{ (ms}^{-1}\text{)}$	A1		(0.34641)
		ii	Use momentum equation	M1		Allow their <i>v</i> ; allow sign errors / omission of <i>m</i>
		ii	$0.3m - 0.6m\cos 30^\circ = am + 0.2\sqrt{3}m\cos 60^\circ$	A1ft	Follow through on <i>v</i>	<i>m</i> 's not necessary;
		ii	(<i>a</i> =) 0.393 to left	A1	Direction must be clearly stated or implied from working. WWW	(0.39282) Away from B/opp direction to before
	Copyright © 2024 Exam Papers Practice	iii	Use of NLR	M1		Allow sign error and / or trig error
		iii	$(0.2\sqrt{3}) \cos 60^\circ - (-0.393) = e (0.6\cos 30^\circ + 0.3)$	A1ft	Ft on <i>a</i> and <i>v</i>	
		iii	0.691	A1	CAO	(0.69082 or 0.6905679)
			Total	9		



Question			Answer/Indicative content	Marks	Part marks and guidance	
21		i	Use of conservation of momentum	M1*	Must be 3 non-zero terms	Allow sign errors, $m/2m$ errors, sin / cos
		i	$2macos\alpha - mbcos\beta = mx2xcos45^\circ$	A1		
		i	Use of NEL	M1*	Must be 3 non-zero terms, and 'e' in correct position	Allow sign errors, sin / cos,
		i	$2cos 45^\circ - 0 = -2/3$ ($-bcos\beta - acos\alpha$)	A1		
		i	Attempt to eliminate $acos\alpha$ or $bcos\beta$	*M1	Dep both previous M1 marks	
		i	$acos\alpha = 5\sqrt{2}/6$	A1	AG	
		i	$bcos\beta = 2\sqrt{2}/3$ oe	A1	Dep final M1 and www	
					Examiner's Comments	
					Almost all candidates used the Law of Conservation of Momentum and Newton's Experimental Law correctly, though there were some sign errors. Most candidates correctly derived the required result, though a small number did not do this convincingly and so did not gain full marks. A small number of candidates also quoted a result gained by using the Law of Conservation of Momentum perpendicular to the line of centres. Usually this was not used, and so was ignored by markers.	
		ii	$asin\alpha = 2$	B1		
		ii	Attempt to solve $asin\alpha = 2$ and $acos\alpha = 5\sqrt{2}/6$	M1	Need to eliminate a or α	
		ii	$a = 2.32$	A1		2.321398...,



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\alpha = 59.5^\circ$	A1	Accept 1.03 radians Examiner's Comments Almost all candidates realised that this required the use of the result quoted in part (i) and as $\sin \alpha = 2$ and efficiently and correctly found the values of a and α . A small number of candidates was unable to make any meaningful attempt at this part. A minority of candidates also worked out b and β .	59.49104...°, 1.0383...rad
			Total	11		



Question			Answer/Indicative content	Marks	Part marks and guidance	
22		i	Realising impulse must be in same direction as velocity, or opposite	M1		
		i	max speed $2.8 \text{ (m s}^{-1}\text{)}$	A1	$0.8 \pm 0.6/0.3$	Various methods
		i	min speed $1.2 \text{ (m s}^{-1}\text{)}$	A1	-1.2 is wrong	
					Examiner's Comments	
					Almost all candidates found the greatest possible speed correctly, and realised that the impulse would act parallel to the particle's original direction of motion. The least possible speed caused some difficulty, with quite a number of candidates giving an answer of -1.2 ms^{-1} , others giving 0, and a small number finding the resultant speed when the impulse acted at right angles to the particle's original direction of motion. A small number of candidates worked with the impulse acting at a general angle to the original direction of motion, and these candidates were usually successful.	
		ii	Impulse momentum diagram	M1	Triangle with sides labelled 0.24, 0.6 and	Allow M1 if positions wrong.
		ii		A1	0.75 or 0.8, 2 and 2.5	Diagram must be correct.
		ii	$\cos \theta = \frac{0.6^2 + 0.24^2 - 0.75^2}{2 \times 0.6 \times 0.24}$	M1		$v_x = 0.8 + 2 \cos \theta$ M1 either $v_y = 2 \sin \theta$ and correct diag A1 both Square, add, giving $1.61 = 3.2 \cos \theta$ M1
		ii	$\theta = 120^\circ \text{ (2.098 rad)}$		Accept $59.8^\circ \text{ (1.04 rad)}$	120.(21)...A1



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	angle shown correctly	A1	Consistent with their θ ; dep M1A1M1 Examiner's Comments Most candidates tackled this part by drawing a triangle with sides of length 0.24, 0.6 and 0.75 (or else 0.8, 2 and 2.5) and then using the cosine rule, though not all put the sides in the right order, and not all found the right angle. However, a significant number of candidates answered the question by resolving momentum, either parallel and perpendicular to the particle's original direction of motion, or else parallel and perpendicular to the impulse. A small number of candidates using this second approach wrongly assumed that only one angle was required in their calculation. Almost all successful candidates clearly showed the required angle, often on their initial sketch.	
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Question			Answer/Indicative content	Marks	Part marks and guidance	
23		i	Using conservation of momentum along loc	M1	3 (or 4) terms, correct dimensions	Allow sign errors, (sin / cos)
		i	$0.1 \times 2.8 + 0.4 \times 1 \times 0.8 = 0.4 \times b$	A1		may see $b = 1.5$
		i	Using NEL	M1	Vel diff after = e x vel diff before	Allow $\pm e$
		i	$b - 0 = -e(1 \times 0.8 - 2.8)$	A1		
		i	$e = 0.75$	A1	Examiner's Comments This part proved one of the most accessible on the paper, with most candidates working out the speed of B along the line of centres after the impact as 1.5 ms^{-1} and then correctly using Newton's experimental law. The most common errors were to omit the mass of B in the term for the momentum of B after impact, failing to use the fact that A was stationary after the impact and using the sum of speeds along the line of centres before impact, rather than the difference, in Newton's experimental law.	
Copyright © 2024 Exam Papers Practice	ii		$b(\text{perp}) = 0.6$	B1		May be on diagram
	ii		$\tan \beta = \frac{b(\text{perp})}{\text{their } 1.5}$	M1*	$\beta = 21.8^\circ$; ft 1.5 from (i)	21.8014...(0.381 rad)
	ii		angle turned through is $36.9^\circ - \beta$	*M1	Must be $36.9^\circ - \text{their } \beta$ (soi)	36.86989



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$= 15.1^\circ$ (0.262 rad)	A1	Examiner's Comments This question was done far better than similar questions have been done recently. Just a few candidates stopped after finding the angle the motion of B made with the line of centres after the impact, and a small number gave the answer as 165° .	15.068 scB1 for 165° after B1M1
			Total	9		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
24		i	$\frac{1}{2}mv^2 = mg \times 0.8(1 - \sin 30^\circ)$	M1	Or with '5m' if for Q	Allow g missing for M1.
		i	$v = 2.8 \text{ m s}^{-1}$	A1		Might see $v^2 = 0.8g$
		i	Speed of P and Q equal	B1ft	soi	
		i	Use conservation of momentum			
		i	$5mx2.8 - mx2.8 = 5mq + mp$	B1ft	Ft on velocity	p is vel of P, q is vel of Q, both to left
		i	Use of NEL	M1		Allow $\pm e$
		i	$p - q = -0.95(-2.8 - 2.8)$	A1ft	Ft on velocity	
		i	$p = 6.3 \text{ m s}^{-1}$	A1	Supporting work required for AG	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	$q = 0.98 \text{ m s}^{-1}$ Q moves to left	A1	Direction must be clear Examiner's Comments Most candidates realised that P and Q would collide at the lowest point of the cylinder, with equal speeds. However, a number of candidates, having realised that the particles would collide at the lowest point, correctly worked out the speed of P, but made an error in the speed of Q, usually through having the mass of Q as m in the expression for KE and $5m$ in the expression for PE. Some follow through marks for the rest of the part were available to these candidates. A small number of candidates worked with the angle that a line joining O to the point of collision makes with the vertical; these candidates generally made no worthwhile progress. A small number of candidates were completely correct, except for not making clear the direction of motion of Q after the collision.	
		ii	By energy for P at top	M1	Must have 3 terms	Allow g missing, sign error
		ii	$\frac{1}{2}m6.3^2 = \frac{1}{2}mv^2 + mg \times 1.6$	A1		
		ii	$v^2 = 8.33$	A1	Soi	
		ii	Use $F = ma$ at top	M1	Must have 3 terms	Allow g missing, sign error
		ii	$mg + R = m \times \frac{8.33}{0.8}$	A1ft	Their v^2	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$R = 0.6125m$	A1CAO	Or $49m/80$ Examiner's Comments This part was done very well. The most common errors were to use the wrong height from the bottom of the cylinder to D , and to miss out the weight term when working out the normal reaction acting on P at D . A small number of candidates worked this out in terms of a general angle, putting that angle equal to 180° , usually correctly, at the end.	
			Total	14		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
25		i	Use of correct formula	M1	$v^2 = 0^2 + 2 \times 9.8 \times 0.4$	or by energy
		i	Vert speed imm before bounce = $2.8 \text{ (ms}^{-1}\text{)}$	A1		
		i	Time between bounces = 0.286 (s) (2/7)	B1	Examiner's Comments Almost all candidates found the velocity correctly, though there were a very small number of scripts where a formula was quoted incorrectly. A small number of candidates omitted to find the time taken for the particle to travel from A to B.	
		ii	Use of their t in a correct formula	M1	$0 = u + 9.8 \times 0.5(t)$ Allow their value of t	or $-u = u - 9.8t$
		ii	Vert speed imm after bounce = $1.4 \text{ (ms}^{-1}\text{)}$	A1		
		ii	Coeff of rest = 0.5	B1ft	Their values for v after / v before Examiner's Comments Almost all candidates did this correctly, though some made an error finding the speed of the particle on leaving A. A small number of candidates worked successfully in terms of a speed of $2.8e$ and did not need to find the speed explicitly.	must be worked out to fraction or decimal; $0 \leq e \leq 1$
		iii	Imp = change of mom	M1	$I = 0.3 \times (v) + 0.3 \times (u)$ Allow their u, v	allow sign errors for M1, allow if answer implies use of their values



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$t = 1.26 \text{ (N s)}$	A1	CAO Examiner's Comments Most candidates gained both marks for this, though some, including those who had earlier errors in speeds, gained only M1.	
			Total	8		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
26		i	Conservation of momentum	*M1	Must have 4 terms Examiner's Comments This question was done extremely well. The errors that did occur were in taking the direction of motion of A in one direction in the momentum equation and the other in Newton's experimental law; in omitting mass from one or more term of the momentum equation or including it in Newton's experimental law, and in making a slip in the solution of the two simultaneous equations.	allow sign errors, $\cos\theta$ omitted
		i		A1	$0.1 \times 3 + 0.2 \times 1 \times \cos\theta = 0.1 \times a + 0.2 \times b$	a and b are vel components of A and B to right, respectively, after collision
		i	Newton's experimental law	*M1	Must have 4 terms and 0.8	allow sign errors, $\cos\theta$ omitted
		i		A1	$b - a = -0.8 (1 \times \cos\theta - 3)$	
		i	Attempt to solve their 2 sim eqns	M1*	Dep both previous M marks	allow 1 slip
		i	0.12 in same direction as before	A1	Direction may be implied by working	withhold if direction stated to left
		ii	$b = 2.04$	B1	Must be seen / used in (ii)	
		ii	vel of B perp to line of centres = 0.8	B1	$(1 \times \sin\theta)$	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	Direction of B after collision makes angle	M1	$\tan \phi = 0.8/2.04$; Examiner's Comments Almost all candidates correctly stated or used 0.8 as the velocity of B perpendicular to the line of centres. Those who had part (i) correct usually arrived at 21.4° for the direction of motion of B after impact, though not all then found the angle turned through correctly. The method mark for finding the direction of B after the impact was available for those who had made errors in part (i), and so was the final followthrough mark for finding the angle turned through.	Allow with their 0.8 and 2.04 (b from (i)); allow $\tan \phi = 2.04/0.8$, if angle clear, leading to 68.4° for A1
		ii	21.4° with line of centres	A1	or 0.374 rads	
		ii	Angle turned through by B is 31.7°	A1ft	or 0.554 rads; allow +/-	53.1 (3) – ϕ , 0.927 – 0.374 rads
			Total	11		



Question			Answer/Indicative content	Marks	Part marks and guidance	
27		i		M1*	Attempt at use of conservation of momentum	
		i	$4(8) + 3(-10) = 4v_A + 3v_B$	A1		
		i	$\frac{1}{2}(4)(8)^2 + \frac{1}{2}(3)(10)^2 - \frac{1}{2}(4)v_A^2 - \frac{1}{2}(3)v_B^2 = 121.5$	M1*		
		i		A1	Attempt at use of KE(before) – KE(after) = 121.5	
		i		M1 dep*	Obtaining quadratic equation in either v_A or v_B ($7v_B^2 - 4v_B - 416 = 0$, $28v_A^2 - 16v_A - 935 = 0$) and attempt to solve quadratic for either v_A or v_B	
		i	$v_A = -5.5$ ($v_A = 6.0714\dots$) so speed of A is $5.5 \text{ (m s}^{-1}\text{)}$	A1	cao; must be positive	
		i	$v_B = 8$ ($v_B = -7.428\dots$) so speed of B is $8 \text{ (m s}^{-1}\text{)}$	A1	cao; must be positive	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	Both particles are moving in the reverse direction to their original motion	A1	Or an equivalent statement consistent with their v_A and v_B; left and right not sufficient without a diagram; moving away from each other needs a diagram also Examiner's Comments A good source of marks for many candidates. A few made sign errors in the two equations but in most cases both the linear and quadratic equation were initially correct. The manipulation to obtain a quadratic in a single v often left much to be desired – errors when dealing with the fractions were abundant and several candidates thought that since $4v_A = 2 - 3v_B$, $4v_A^2$ could be replaced by $(2 - 3v_B)^2$. The necessity for answers to be positive was often not recognised, directions were sometimes indicated by arrows rather than a statement and statements, when given, more often than not included reference to left / right, or moving apart rather than the more explicit 'reverse direction to original'.	
		ii	$v_A - v_B = -e(8 - (-10))$	M1	Attempt at use of coefficient of restitution, right way round, v_A and v_B substituted	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$e = 0.75$	A1	Examiner's Comments The majority of candidates, who found a solution to part (i), were able to use the restitution equation, with only a few sign errors seen due to not taking into account the vector nature of the values used.	
			Total	10		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
28		i	$v^2 = 5^2 + 2g(1.6)$	B1	Complete method to find v (= 7.507...)	
		i	$0.7 \times 7.507... (= 5.255...)$	B1	$0.7 \times cv(v)$, but not $cv(v) = 5$; may be seen in (ii)	
		i	$(0.7 \times cv(v))^2 = 2gh$	M1	Complete method to find h , with final speed 0; allow $cv(v) = 5$ for method	
		i	$h = 1.41 \text{ m}$	A1	Exact 1.409 Examiner's Comments The vast majority of candidates scored full marks here. Of those who didn't, the common error was to use 5 m s^{-1} as the speed when hitting the ground.	
		ii		M1	Change in momentum found, with relevant velocities ie $cv(v)$ and $0.7 \times$ $cv(v)$ but not $cv(v) = 5$	
		ii	$0.2(7.507...)(0.7) - (-0.2)(7.507...)$	A1ft	This may be negative; ft on their v found in (i)	
		ii	Impulse = 2.55 N s, upwards	A1	(2.5524...) Must have direction also. Examiner's Comments Many candidates who were able to answer (i) correctly went wrong here. The common error was not to deal correctly with the vector aspect of the momentum of the sphere with a difference of two scalar quantities being seen significantly often.	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii		M1	Change in energy found, all energy terms needed and no extra terms This may be negative	
		iii	$0.2(9.8)(1.6) + \frac{1}{2}(0.2)(5^2) - 0.2(9.8)(cv(h))$ OR $\frac{1}{2}(0.2)(7.507\dots^2) - \frac{1}{2}(0.2)(0.7 \times 7.507\dots)^2$	A1ft		
		iii	Loss of energy = 2.87 J	A1	(2.87436 exact); art 2.87; allow -2.87 Examiner's Comments This was done relatively poorly. The problem was either through not considering all the energy terms needed (ignoring PE or KE) or using far too many (bringing in the 7.507 and/or the 5.255).	
			Total	10		



Question			Answer/Indicative content	Marks	Part marks and guidance	
29		i		M1	Attempt at use of N2L with $a = v^2/r$ or $a = r\omega^2$; allow verification.	
		i	$0.4v^2/0.6 = 6, v = 3$	A1	AG If ± 3 then -3 must be rejected for A1. Examiner's Comments Not all candidates recognised that since the answer was given there was a need for them to show their working clearly. There were many good solutions but a disappointing number wrote a formula and then gave a value for ω or v^2 without showing the values used to obtain it. A few candidates did not know how to proceed and seemed to think the coefficient of restitution was relevant in this part.	
		ii		*M1	Attempt at use of conservation of linear momentum	
		ii	$3(0.4) = 0.4v_A + 0.5v_B$	A1		
		ii		*M1	Attempt at use of restitution equation, must be correct way round	
		ii	$v_B - v_A = 3(0.35)$	A1	Must be consistent with the directions used for CoLM	
		ii		dep*M1	Solving simultaneous equations	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$v_A = 0.75 \text{ m s}^{-1}$ and $v_B = 1.8 \text{ m s}^{-1}$	A1	Both values positive as final answer. Examiner's Comments The majority of candidates scored full marks on this part. For those who guessed the wrong direction for one of their velocities, the issue was resolved at the end and speed was stated successfully.	
		iii	$Y = 0.4g$ (= 3.92)	B1	Resolve vertically	
		iii	$(4.9)^2 - (0.4g)^2$ (= 2.94 ²)	B1	Use of Pythagoras	
		iii	$0.4(0.6)\omega^2 = cv(2.94)$	M1	Resolve horizontally, $cv(2.94)$ from a legitimate attempt at 2.94	
		iii	$\omega = 3.5$	A1	Exact	
		iii	OR $4.9\cos\theta = 0.4g$	B1	Resolve vertically, θ angle with vertical	
		iii	$\cos\theta = 0.8$ or $\theta = 36.9$	B1		
		iii	$0.4(0.6)\omega^2 = 4.9\sin\theta$	M1	Resolve horizontally, θ substituted and from a legitimate attempt	
		iii	$\omega = 3.5$	A1	Exact	
		iii	OR $Y = 0.4g$ (= 3.92)	B1	Resolve vertically	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$X = 0.4(0.6)\omega^2 (= 0.24\omega^2)$	B1	Resolve horizontally	
		iii	$(0.4g)^2 + (0.4(0.6)\omega^2)^2 = 4.9^2$	M1	Attempt at use of Pythagoras, from a legitimate attempt at X	
		iii	$\omega = 3.5$	A1	Exact	
					Examiner's Comments	
					This part was poorly done by a significant majority of candidates, who were unable to find the force required to proceed. Many thought that 4.9 N was the force towards the centre. Other common errors seen were for the force towards the centre to $(4.9 - 0.4g)$ N, as well as some who thought the 4.9 N force was at an angle of 45° to the horizontal.	
			Total	12		



Question			Answer/Indicative content	Marks	Part marks and guidance	
30		i		*M1	Attempt at use of conservation of momentum	
		i	$(2m)(4) - (3m)(2) = 2mv_A + 3mv_B$	A1		
		i		*M1	Attempt at use of coefficient of restitution	
		i	$(v_B - v_A)/(4 - -2) = 0.4$	A1		
		i		Dep**M1	Solving for v_A and v_B	
		i	Speed A = 1.04 m s^{-1} , Speed B = 1.36 m s^{-1}	A1	Final answers must be positive	
					Examiner's Comments	
					Most candidates know the standard approach to this type of request. Using conservation of momentum and Newton's experimental law to produce two simultaneous equations was usually successfully done, with only a minority of candidates using speed rather than velocity in these two equations. Unfortunately 5 marks were almost as common as full marks, due to candidates giving the velocities and not the speeds as their final answer.	
		ii	Energy before = $\frac{1}{2}(2m)(4^2) + \frac{1}{2}(3m)(2^2)$	B1ft	Energy before or Loss in A's KE	
		ii	Energy after = $\frac{1}{2}(2m)(1.04^2) + \frac{1}{2}(3m)(1.36^2)$	B1ft	Energy after or Loss in B's KE	
		ii	$22m - 3.856m$	M1	Difference of total OR sum of differences (total kinetic energy must decrease)	
		ii	$18.1m$	A1	$18.144m$ (Exact)	



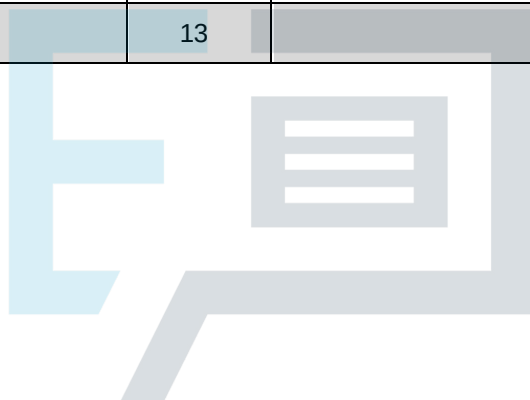
Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	OR $\frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} (1 - e^2) A^2$	*B1	Loss of kinetic energy formula, where A = approach speed	
		ii		Dep*M1	Substitution of values into quoted formula	
		ii	$\frac{1}{2} \frac{(2m)(3m)}{2m + 3m} (1 - 0.4^2) (4 + 2)^2$	A1		
		ii	18.1m	A1	18.144m (Exact)	
					Examiner's Comments	
					Many fully correct answers or scores of 3/4 obtained with follow through. Some carelessness with the masses as m instead of $2m$ and $3m$ was seen. Also in some instances the m had a tendency to disappear at the subtraction stage; in some cases, but not all, it reappeared in the final answer.	
		iii		M1	Attempt at change in momentum and equate to impulse. Must use $2m$ or $3m$	
		iii	$2m(4) - 2m(-1.04) = 2.52$	A1ft	Or $3m(2) - 3m(-1.36) = 2.52$	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$m = 0.25$	A1	Exact Examiner's Comments This was the least well done part of this question. Errors in signs often leading to a negative mass, with the sign ignored at the final stage, were often seen. Confusion about what mass to use was significant. The mass was frequently taken to be m , in a few cases $5m$ was used or a mixture of $2m$ and $3m$ used in the same equation.	
			Total	13		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
31		i		M1	Use of restitution, may have sign errors, must be correct ratio (v/u)	
		i	$4 - 4(1 - e + e^2) = -e(u - 4)$	A1		
		i	$u = 4e$	A1	oe	
		i		M1	Use of conservation of momentum	
		i	$mu + 0.2 \times 4 = 0.2 \times 4(1 - e + e^2) + 4m$	A1		
		i	$m = 0.2e$	A1	oe	
					Examiner's Comments	
					Most candidates knew which principles to apply, and their attempts to apply them usually gained marks. The momentum equation usually began correctly, though the coefficient of restitution equation was less reliably correct. A major source of error was in multiplying out brackets. Many failed to distribute the 4 to all terms when expanding $4(1 - e + e^2)$; $e(4 - u)$ often became $4e - 4u$.	
		ii	Valid method to find e that gives the least speed	M1	Differentiate v_A and equate to 0 or complete the square on v_A	
		ii	Get $e = \frac{1}{2}$	A1	www	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii		M1	Difference of KE with 4 terms	
		ii	$\frac{1}{2} \times 0.2 \times 4^2 + \frac{1}{2} \times 0.1 \times 2^2 - (\frac{1}{2} \times 0.2 \times 3^2 + \frac{1}{2} \times 0.1 \times 4^2)$	A1	Must have found the value of e from a legitimate method. www	
		ii	(+/-) 0.1 J	A1	SCM1A1 Loss of KE = $8e(1 - e)^3/5$ or $8e(1 - 3e + 3e^2 - e^3)/5$ or $8e/5 - 24e^2/5 + 24e^3/5 - 8e^4/5$ Examiner's Comments Many good arguments were presented leading to $e = \frac{1}{2}$, using completing the square, the discriminant of $e^2 - e + (1 - k) = 0$, or differentiation. As the earlier parts of the question had presented so many opportunities for error, the correct answer to the loss in kinetic energy was not usually obtained. Many had a good effort with KE, though often only one particle was considered.	
	Copyright © 2024 Exam Papers Practice	iii		*M1	Attempt to use impulse = change in momentum on one particle	
		iii	$0.2e(4 - 4e) = 0.192$ or $0.2(4 - (4 - 4e + 4e^2)) = 0.192$	A1		
		iii	Solve three term QE in e ,	dep*M1	Method should lead to 2 real values for e	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$e = 0.4$ or 0.6	A1	For both Examiner's Comments Those who chose to deal with particle A, maximised their chances of gaining the full four marks since they were not relying on their own answers from part (i). Sign errors were common which lead to answers of $e = 1.2$ and -0.2 , both of which should have been a clue that something had gone wrong in this part. Those who had not simplified their answers for m and u frequently found higher order equations to solve for e .	
			Total	15		



Question			Answer/Indicative content	Marks	Part marks and guidance	
32		i	$a = g \sin 30$	B1		
		i		M1	Using NEL with u_A from $cv(a)$, u_A	
		i	$1 + u = 0.4(2 + 2g \sin 30)$	A1	$\neq 0$	
		i	$u = 3.72 \text{ ms}^{-1}$	A1	cwo	
					Examiner's Comments Setting the motion on an inclined plane caused problems for a significant number of candidates. Solutions in which the speed of A before the collision was taken to be zero or found from using an acceleration of g were all too common. Many were able to use the coefficient of restitution correctly with only a few examples of wrong signs with velocities being used.	
		ii	Use $v^2 = u^2 - 2(g \sin 30)s$	M1	Using $v = 0$, $cv(a)$ from (i) or	
		ii	$s = 1.41 \text{ m}$	A1	correct a SC If a not found in (i), allow $a = g$ for M1A0.	
					Examiner's Comments The usual approach seen was to use constant acceleration equations using their acceleration found in (i). It was surprising though that a significant number, having found acceleration as $g \sin 30$ in (i), used g in this part.	
		iii	Use of conservation of momentum	M1	Using $cv(a)$	
		iii	$0.5 \times 2g \sin 30 - 2m = m - 0.5 \times 3.72$	A1ft	ft $cv(u)$ from (i)	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$m = 2.25$	A1	Aef(raction) eg $2^{19/75}$ or $169/75$ Examiner's Comments Most recognised the need to use conservation of momentum to solve this request and the signs of the individual terms were usually correct. A few candidates used the masses with the wrong velocities.	
			Total	9		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
33		i	Use $I = mv$	M1		
		i	3.6 ms^{-1}	A1	–3.6 gets A0 Examiner's Comments This proved to be the most accessible question on the paper. Only a small minority were unfamiliar with the relationship between impulse and momentum.	
		ii	$\pm \frac{1}{2} \times 0.5 \times 3.6^2 - \frac{1}{2} \times 0.5 \times v^2$	B1		
		ii	$0.5 \times g \times 0.3$	B1		
		ii	Use of conservation of energy	M1	Three terms	
		ii	$v = 2.66 \text{ ms}^{-1}$	A1		
		ii	OR $a = -g \sin \theta$	B1	θ angle of plane to horizontal	
		ii	$s = 0.3 / \sin \theta$	B1		
		ii	Use $v^2 = u^2 + 2as$	M1	$a \neq -g$, $s \neq 0.3$.	
		ii	$v = 2.66 \text{ ms}^{-1}$	A1	Examiner's Comments Most candidates used the intended energy approach. This was the most successful method with only a minority having the change in kinetic energy with a wrong sign. It was possible to use constant acceleration to solve the problem if a general angle of the slope was used. Unfortunately many, wrongly, used acceleration as g and displacement as 0.3 m.	
		iii		M1	Difference of KE and PE	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	Change in energy = $\pm(\frac{1}{2} \times 0.5 \times 3^2 - 0.5 \times g \times 0.2)$	A1		
		iii	Equate to force x distance	M1	Attempt at $0.2/\sin 30$ for dist, 3 terms	
		iii	3.175 N	A1	Allow 3.18	
		iii	OR Using $v^2 = u^2 + 2as$ to find a	M1		
		iii	Resolve parallel to plane	M1	Use $v = 0$, attempt at $s = 0.2/\sin 30$	
		iii	$0.5g\cos 60 + F = 0.5 \times cv(11.25)$	A1	N2L used with cv(11.25), 3 terms Consistent signs	
		iii	$F = 3.175$	A1	Allow 3.18	
					Examiner's Comments	
					Two common methods were seen by examiners. The energy approach was usually well done, with common errors seen as using conservation of energy with wrong signs and also the potential energy of the particle used twice. The distance of $0.2/\sin 30$ m in the work done by the frictional force was usually attempted. Candidates, who used constant acceleration combined with Newton's second law, commonly thought the displacement to be used was 0.2 m rather than $0.2/\sin 30$ m.	
			Total	10		



Question			Answer/Indicative content	Marks	Part marks and guidance	
34		i	$0.8 \times 6 - 0.2 \times 2 (=4.4)$	B1	Before momentum, signs different, no g	
		i		M1	Uses momentum conservation, no g	
		i	$0.8 \times 6 - 0.2 \times 2 = 0.8 \times 4 + 0.2v (= 4.4)$	A1		
		i	$v = 6 \text{ m s}^{-1}$	A1	B 's "after" velocity Examiner's Comments This was almost always answered correctly.	
		ii	After mass = $0.3 + 0.1$	B1		
		ii	$0.3 \times 5 (+0.1 \times 0) = (0.3 + 0.1)v$	M1	No g	
		ii	$v = 3.75 \text{ m s}^{-1}$	A1	CD "after" velocity Examiner's Comments This was correctly answered even more often than part (i).	
		iii	Least final speed $B = 4$	B1	It cannot be less than the speed of A	
		iii	$0.2 \times 6 + (0.3 + 0.1) \times 3.75 = 0.2 \times (v \geq 4) + 0.4V$	M1	Momentum, B and CD particles, essentially 4 non-zero terms with distinct velocities. Letters used at this stage should be checked against values used later.	
		iii	$0.2 \times 6 + (0.3 + 0.1) \times 3.75 = 0.2 \times 4 + 0.4V$	A1✓	ft cv ($v(i)$ and $v(ii)$)	



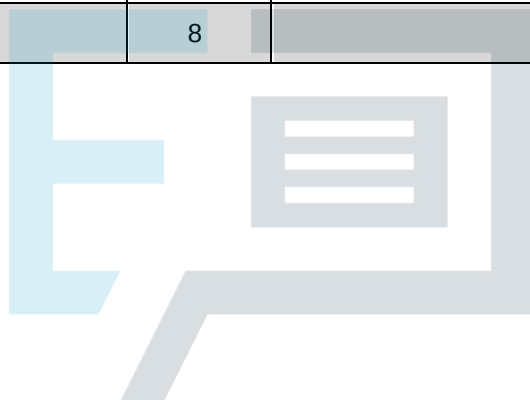
Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$V = 4.75 \text{ m s}^{-1}$	A1	Examiner's Comments This was rarely answered well, using the understanding that the speed of <i>B</i> after the third collision must be at least 4 ms^{-1} if it is not to be struck again by <i>A</i> . The most common assumption was that <i>B</i> would come to rest if the combined particle <i>C/D</i> were to have its greatest speed. The second most common assumption was that <i>B</i> and <i>C/D</i> would coalesce.	
			Total	11		



Question			Answer/Indicative content	Marks	Part marks and guidance	
35		i	Before momentum = $\pm$ (0.4u – 0.3 × 8)	B1		Accept inclusion of <i>g</i> , including final A1
		i		M1	Uses momentum cons. 4 non-zero terms	
		i	0.4u – 0.3 × 8 = – 0.4u + 0.3 × 8	A1ft	ft candidates “before” expression	
		i	u = 6	A1	Examiner's Comments There were many excellent solutions to this question, although some candidates showed the usual weakness when dealing with the positive and negative signs required to indicate the directions of the particles. The most common error in (i) was failure to consider conservation of momentum of the whole system.	
		ii	After momentum = $\pm$ 9m	B1		No marks if <i>g</i> included, even if apparently cancelled
		ii		M1	Uses momentum conservation 3 non-zero terms	
		ii	0.3 × 8 – 3m = 9m	A1ft	ft candidates “after” expression	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$m = 0.2$	A1	Examiner's Comments There were many excellent solutions to this question, although some candidates showed the usual weakness when dealing with the positive and negative signs required to indicate the directions of the particles. In (ii) a small minority of candidates were unable to give the correct solution to the equation $12m = 2.4$ but, if they reached this point, only lost the final mark.	
			Total	8		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
36		i	Calculation for both “before” Momentum (magnitudes)	M1	Must not include g	Explicit reference to after momentum or conservation of momentum essential.
		i	Compares both terms without arithmetic error	A1*		
		i	Shows direction of after total momentum conflicts with the before velocity / momentum of Q	D*A1	Vector nature of momentum by word or sign (+/-) Examiner's Comments The explanation rested on the basic facts that the total “after” momentum has a sense which is in conflict with the “before” motion of Q. Convincing reasons for the change in the direction of motion of Q came from candidates who based their case on a form of proof by contradiction.	
		ii	$TMB = +/-(0.2 \times 4 + 0.3 \times (-2))$	B1	Accept inclusion of g	LHS must be difference for both M1 marks
		ii	$0.8 - 0.6 = 0.2v + 0.3v$	M1	Allow if g included in all terms	
		ii	$v = 0.4 \text{ ms}^{-1}$	A1	Not awarded if g included	
		ii	$0.8 - 0.6 = -0.2v + 0.3v$	M1	Allow if g included in all terms	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$v = 2 \text{ ms}^{-1}$	A1	Not awarded if g included Examiner's Comments Was often answered well, though some solutions were based on equations implying that Q would continue to move in its old direction after the collision. Candidates who offered only one solution used the case where the two particles effectively coalesce.	Speed = 2 and the direction of motion of Q is reversed A1
			Total	8		



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